

Math 465: Homework 4

Due: Tuesday, September 29, by 11:59PM

Instructions. Write full and complete solutions to the following problems and submit them on Gradescope by the deadline. For each problem, explain the reasoning behind your answer; a formula or numerical answer without justification is not a complete solution. **If your final answer is numerical, please carry out the arithmetic** and state the resulting number (so, for example, if your answer is $6! - 5!$, I want you to also tell me the answer is 600.)

You may discuss the problems with other students, but you must write up your solutions independently and in your own words. You are not allowed to use generative AI tools, Internet searches, or any other external means to directly obtain a solution. Generative AI tools may also not be used to proofread your writing or L^AT_EX. If you are unsure about what is or is not permitted, please ask.

At the beginning of your submission, acknowledge any permitted books, notes, or other sources you consulted, and list the names of any students with whom you discussed the homework.

1. Give a combinatorial proof of the recurrence

$$c(n, k) = c(n-1, k-1) + (n-1)c(n-1, k) \quad (n \geq 2, 1 \leq k \leq n),$$

where $c(n, k)$ is the signless Stirling number of the first kind. Use the conventions $c(m, 0) = 0$ for $m \geq 1$ and $c(m, k) = 0$ for $k > m$.

2. Give combinatorial proofs of the following identities. Here $c(n, k)$ and $S(n, k)$ are the signless Stirling numbers of the first kind and the Stirling numbers of the second kind, respectively.

- (a) $c(n, n-1) = \binom{n}{2}$ for $n \geq 2$.
- (b) $c(n, n-2) = 2\binom{n}{3} + 3\binom{n}{4}$ for $n \geq 3$.
- (c) $S(n, n-2) = \binom{n}{3} + 3\binom{n}{4}$ for $n \geq 3$.

3. A group of n distinct tourists arrive at a restaurant. They sit around an unspecified number of nonempty circular tables, and each table orders one of r possible drinks, where $n, r \geq 1$. Tables are unlabeled, and two seating arrangements are considered the same if every person has the same left neighbor in both. Prove that the number of possible seating-and-drink arrangements is

$$r(r+1) \cdots (r+n-1).$$

4. Give a combinatorial proof that, for integers $0 \leq k \leq n$,

$$\sum_{j=k}^n \binom{j}{k} = \binom{n+1}{k+1}.$$

Do not use Pascal's recurrence or algebraic manipulation of binomial coefficients.

5. Find a closed formula for

$$\sum_{j=0}^n j^3 \quad (n \geq 0).$$

6. Solve each of the following recurrences, giving a closed formula for a_n for every $n \geq 0$.

(a)

$$a_n = 2a_{n-1} + a_{n-2} - 2a_{n-3} \quad (n \geq 3),$$

with initial values $a_0 = 4$, $a_1 = 8$, and $a_2 = 10$.

(b)

$$a_n = 9a_{n-1} - 27a_{n-2} + 27a_{n-3} \quad (n \geq 3),$$

with initial values $a_0 = 1$, $a_1 = 3$, and $a_2 = 27$.

7. Find a third-order homogeneous linear recurrence with constant coefficients satisfied by the sequence

$$a_n = (2n - 1)3^n + 4(-1)^n \quad (n \geq 0).$$

You do not need to specify initial conditions.

8. Recall that the Fibonacci numbers are defined by $f_0 = 0$, $f_1 = 1$, and $f_n = f_{n-1} + f_{n-2}$ for $n \geq 2$. Give a combinatorial proof that

$$f_{n+1} = \sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n-k}{k} \quad (n \geq 0).$$

9. Give a combinatorial proof that, for all integers $m, n \geq 1$,

$$f_{m+n+1} = f_{m+1}f_{n+1} + f_m f_n.$$

10. For $n \geq 1$, let a_n be the number of compositions of n in which every part is odd.

(a) Starting from the Fibonacci recurrence and initial conditions, find a closed rational expression for

$$F(x) = \sum_{n \geq 0} f_n x^n.$$

(b) Find a closed rational expression for

$$A(x) = \sum_{n \geq 1} a_n x^n,$$

and use it to express a_n in terms of Fibonacci numbers.

(c) Give a bijection between compositions of n into odd parts and compositions of $n - 1$ into parts equal to 1 or 2. Explain why your bijection proves the formula obtained in part (b).